Spatiotemporal Data Fusion of Index-Based VTCI Using Sentinel-2 and -3 Satellite Data for Field-Scale Drought Monitoring

Yue Zhang, Pengxin Wang, Kevin Tansey, Mingqi Li, Fengwei Guo, Junming Liu, and Shuyu Zhang

***Abstract*— Due to climate change, the impact of drought on field crop production is extremely important. This study focuses on the vegetation temperature condition index (VTCI), an index-based drought monitoring index that can characterize drought conditions in near real time (at ten-day intervals), and explores the applicability of different spatial and temporal data fusion schemes to it. It also proposes a field-scale VTCI fusion framework based on the Sentinel-3 VTCI calculation and the land surface temperature (LST) downscaling. First, based on analyzing the computational characteristics of VTCI, multiyear VTCI based on Sentinel data sources was obtained, which further expands the diversity of data sources for VTCI. On this basis, a combination of qualitative and quantitative methods was used to compare the applicability of two schemes: Scheme 1, based on the "blend-then-index" (BI) strategy, which first fuses normalized difference vegetation index (NDVI) and LST, and then calculated the fused VTCIs; and Scheme 2, based on the "index-then-blend" (IB) strategy, which directly fuses the VTCIs based on the calculated VTCIs. It was found that all the fused VTCIs remained highly correlated with the ten-day cumulative precipitation. Compared with the fused VTCIs obtained by Scheme 2, the VTCIs obtained by Scheme 1 were able to display more spatial details. In addition, the VTCIs of Scheme 1 were more consistent with the Sentinel-3 VTCIs, and the accuracy of field yield estimation using the fused VTCIs was higher (r of 0.58 and root-mean-square error (RMSE) of 783.27 kg/ha).**

***Index Terms*— Drought monitoring, field scale, Sentinel-2/3, spatiotemporal data fusion, vegetation temperature condition index (VTCI).**

I. Introduction

DROUGHT is one of the greatest threats to sustainable development, with its occurrence having increased notably since the year 2000. According to data from the United Nations Office for Disaster Risk Reduction (UNDRR), the number and duration of droughts have increased by 29% compared with the previous two decades, with widespread and cascading impacts on societies, ecosystems, and economies [1], [2]. Among them, agricultural drought, as the type of drought that directly affects agricultural production, reflects the extent to which soil moisture falls below the minimum crop water requirements by analyzing soil moisture and morphological characteristics during crop growth [3], [4]. Agricultural droughts occur at different stages of growth and have different impacts on agricultural production. For example, the greatest amount of water is needed during the growing season and when the crop is ready for jointing or flowering, so if drought occurs during this period, it can have a serious impact on crop yields [5]. Therefore, timely and accurate drought monitoring plays an extremely important role in guaranteeing food security, formulating rational production plans, and taking timely adjustment measures.

The complexity of the drought phenomenon itself makes direct observation of its onset time, evolutionary process, and the extent of its impact challenging, and thus, drought indices are often employed to identify and characterize drought events [6]. Currently, there are hundreds of indices to characterize drought, which can be specifically classified into two categories: indices based on ground observation data and remote sensing data [7]. Among them, the method based on ground observation data uses meteorological, soil, and vegetation information obtained from ground stations for drought monitoring, including indices, such as the Palmer drought severity index (PDSI) [8], standardized precipitation index (SPI) [9], China-Z index (CZI) [10], and standardized precipitation evapotranspiration index (SPEI) [11] to name a few. Although indices derived from ground-based observations can provide accurate parameters of drought conditions, the existence of spatial and temporal discontinuities and the need to deploy a large number of stations for their regional or even global application, which, in turn, incur high research costs, are drawbacks that further limit indices based on ground observation data development [12], [13]. In contrast, the remote sensing-based monitoring method has the advantage of spatial continuity, which allows long-term monitoring of large areas and low monitoring costs and has been widely used for drought monitoring. For example, in terms of crop growth characteristics, normalized difference vegetation index (NDVI) [14] and leaf area index (LAI) [15] can provide information on the degree of vegetation cover and growth

Manuscript received 14 September 2023; revised 1 November 2023; accepted 23 November 2023. Date of publication 1 December 2023; date of current version 13 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42171332; and in part by the Royal Society-Newton Mobility Grant, U.K. *(Corresponding author: Pengxin Wang.)*

Yue Zhang, Pengxin Wang, Mingqi Li, and Fengwei Guo are with the College of Information and Electrical Engineering and the Key Laboratory of Agricultural Machinery Monitoring and Big Data Applications, Ministry of Agriculture and Rural Affairs, China Agricultural University, Beijing 100083, China (e-mail: yuezhang_1920@163.com; wangpx@cau.edu.cn; lmq0508@cau.edu.cn; fengwei_guo@163.com).

Kevin Tansey is with the School of Geography, Geology and the Environment, University of Leicester, LE1 7RH Leicester, U.K. (e-mail: kjt7@leicester.ac.uk).

Junming Liu is with the College of Land Science and Technology, China Agricultural University, Beijing 100193, China (e-mail: liujunming2000@163.com).

Shuyu Zhang is with the Shaanxi Provincial Meteorological Bureau, Xi'an 710014, China (e-mail: 985341311@qq.com).

Digital Object Identifier 10.1109/TGRS.2023.3338623

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

status to help assess crop sensitivity to drought and stress levels. From the consideration of crop growth environment, land surface temperature (LST) [16] is often used to assess the thermal condition of the ground and the evaporation capacity of water, thus enabling drought monitoring studies. It is necessary to recognize that drought, as a complex hazard, is caused by the interaction of many factors. Relying on a single monitoring index will fail to meet the requirements for comprehensive drought monitoring. Instead, the combination of multiple indices provides an effective research approach to improve the accuracy of drought monitoring. Based on the consideration that NDVI and LST scatter plots over a region were triangularly distributed, Wang et al. [17] proposed the vegetation temperature condition index (VTCI) combining NDVI and LST to compensate for the deficiency of a single remotely sensed parameter and to provide an index reflecting drought information based on temperature and vegetation [18]. The most critical aspect of the VTCI calculation is the determination of the warm and cold edges. Compared with the VTCI obtained by using only single-year data to determine the warm and cold edges to characterize the relative wet and dry conditions, Sun et al. [19] further used multiyear NDVI and LST data at a ten-day interval to determine the warm and cold edges, so that the calculated VTCI can ensure the comparability of drought among different years and achieve further improvement of the VTCI. In recent years, the VTCI has been effectively employed for drought monitoring in various regions, such as the Guanzhong Plain of China [20], the Great Plains of the United States [21], and the Punjab region in Pakistan [22], [23]. However, recent research has found that the limited temporal and spatial resolution would increase the uncertainty of regional or field drought monitoring when using drought indices [7]. Therefore, how to effectively improve the temporal and spatial resolution to reduce the uncertainty of drought monitoring has become a research topic worthy of attention.

The continuous development of sensor technology provides an effective solution to further reduce the uncertainty of drought monitoring results due to insufficient resolution. At the regional scale, remote sensors, such as advanced very high-resolution radiometer (AVHRR) and moderate-resolution imaging spectroradiometer (MODIS) can provide time-series drought monitoring results from 250 to 1000 m. On the field scale, due to the existence of mixed pixels, it is necessary to utilize Landsat and other sensors to provide drought monitoring results with 30-m spatial resolution that are similar to the field scale in China while clearly distinguishing between different feature types. However, the temporal resolution of these sensors for field monitoring is usually more than ten days, and the data obtained on clear, cloudless days are more limited, which further impacts the accuracy of drought monitoring. The Sentinel series of Earth observation satellites, developed and operated by the European Space Agency (ESA), have a variety of sensors that target different mission objectives to meet the needs of various domains. In particular, Sentinel-2 is aimed at providing high-resolution multispectral imagery for terrestrial monitoring to obtain image information, which includes vegetation, soil, water cover, inland waters, and coastal areas. Sentinel-2 consists of two satellites, A and B, both carrying multispectral imager (MSI). Each satellite has a revisit period of ten days, and the two satellites complement each other with a revisit period of five days. The data contain 13 wavelength bands from visible to short-wave infrared, with spatial resolutions covering 10, 20, and 60 m and an image width of 290 km, allowing for high-resolution ground-based observations. It is crucial to emphasize that of the available optical data, and the Sentinel-2 data are currently the only one that covers the range of the three red-edge bands, which is particularly valuable for monitoring information on vegetation health. The mission of Sentinel-3 is to obtain information on ocean surface topography, LST, and sea and land surface color through high-precision and high-reliability measurements in support of ocean forecasting systems, environmental monitoring, and climate monitoring. The Sentinel-3 series of satellites carry different types of sensors with spatial resolutions of both 300 and 1000 m and a temporal resolution of one day, making them suitable for detailed and frequent monitoring of surface features and changes. However, despite the successful development of various remote sensing satellites, it is still not possible to obtain both high spatial resolution and high temporal resolution remote sensing images due to the limitations of current sensors. The temporal and spatial data fusion method is currently one of the effective methods for solving the above problem, which involves fusing data in the temporal and spatial domains, utilizing high spatial resolution data to obtain spatial detail information while taking advantage of high temporal resolution data to capture information on the evolution of spatial information over time. By efficiently processing the high spatial low temporal resolution data and high temporal low spatial resolution data, data with high spatial and temporal resolution can be generated, thus realizing the "prediction" of the target date. Currently, a variety of spatiotemporal data fusion methods have been developed [24]. Among them, the spatial and temporal adaptive reflectance fusion method (STARFM) proposed by Gao et al. [25] is the first used spatiotemporal data fusion method. In an attempt to better simulate the spatial variability, Zhu et al. [26] proposed an enhanced STARFM (ESTARFM) based on the STARFM method. To further complement the description of spatiotemporal variation features, Zhu et al. [27] proposed the flexible spatiotemporal data fusion (FSDAF) algorithm, which first obtained the temporal feature variation by image element decomposition, and introduced the thin plate spline (TPS) interpolation method to indicate the variation of spatial features, which was used to complement the temporal feature variation to obtain the final fusion results. Unlike other fusion methods, STARFM uses only one pair of reference images for fusion studies, which is suitable for homogeneous regions, such as large agricultural fields. Therefore, with the advantage of the data quantity condition and the specificity of regional features, this method has been widely used in the application of crop growth monitoring [28] and crop information extraction [29].

When using the spatiotemporal data fusion method to fuse the vegetation indices, according to the order of index calculation and data mixing, the fusion strategies can be divided

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

into two categories: "blend-then-index" (BI) and "index-then-blend" (IB) [30], [31]. For the BI method, the reflectance bands are fed into the blend algorithm to simulate reflectance, which, in turn, is used to calculate the multispectral index. For the IB method, the indices are first calculated and then fused using a blend algorithm. Many studies have also explored further the applicability of the fusion strategy, e.g., Jarihani et al. [30] explored the applicability of the two strategies for indices commonly used in vegetation studies, environmental moisture assessment, and standing water identification, and the results showed that the IB approach produced higher accuracies than the BI approach for both blending algorithms for all nine indices. Based on the study of Jarihani et al. [30], Chen et al. [32] further considered the effectiveness of the fusion strategy of using STARFM to fuse NDVI during different crop growth stages. The results showed that the STARFM could generate high-resolution NDVI when the NDVI value at the prediction date was higher than the input NDVI value, and the error of BI was less than IB and vice versa. Although numerous studies have been conducted on the applicability of different strategies, the focus has been mainly on the reflectance-to-index computation level. The applicability of fusion strategies for indices obtained based on multiple index calculations, e.g., VTCI is based on NDVI and LST calculations, has not been adequately investigated. Therefore, there is a need for a targeted exploration of the applicability of fusion strategies for index-based indices.

Although the research on spatiotemporal data fusion has achieved fruitful results, there are still the following problems in applying it to drought monitoring: 1) most of the data used for fusion are MODIS and Landsat data, and the diversity of the data sources needs to be further improved, and 2) when utilizing fusion to achieve drought monitoring, the fusion research based on VTCI is relatively less, and at the same time, since the calculation formula of VTCI involves two variables, NDVI and LST, the kind of fusion scheme that is more conducive to the improvement of the accuracy of the fusion and the effectiveness of the drought monitoring still needs to be further explored.

Based on the above problems, this study has the following objectives: 1) use Sentinel series data for VTCI calculation and verify the validity of Sentinel source VTCI; 2) build two fusion schemes to explore the applicability of BI and IB strategies in fusing index-based indices, i.e., VTCI obtained based on NDVI and LST; namely, Scheme 1, based on the BI strategy, first fused NDVI and LST, and then calculated the fused VTCI; Scheme 2, based on the IB strategy, directly fused VTCI based on the calculated VTCI; and 3) demonstrate the effectiveness of VTCI fusion schemes and field-scale drought monitoring by using a combination of qualitative and quantitative approaches.

II. Study Area and Data

A. Study Area

The Guanzhong Plain, located in central Shaanxi Province, China, was formed by the alluvial deposit of the Wei River. The geographical location is 106°22′E–110°24′E and 33°57′N–35°39′N, as shown in Fig. 1, with an east–west width of about 360 km, a north–south length of about 80 km, an average elevation of about 500 m, a high terrain in the west and a low terrain in the east, and a total area of about 3.6×10^4 km^2. The climate of the Guanzhong Plain is a temperate continental monsoon climate, with an average annual temperature of 6 °C–13 °C and an average annual precipitation of 500–700 mm, which provides good water and heat conditions. The Wei River runs from west to east, providing favorable irrigation conditions and suitable for crop cultivation, making it an important grain producing area in China. The main type of cultivation in the Guanzhong Plain is to rotate summer maize and winter wheat in irrigated areas and to cultivate winter wheat in rainfed areas. Winter wheat is usually sown in early or mid-October and harvested in early June of the following year. The growth stages of winter wheat after overwintering can be categorized into the green-up stage (from early March to mid-March), jointing stage (from late March to mid-April), heading–filling stage (from late April to early May), and milky maturity stage (from mid-May to late May), and these four growth stages are referred to as the main growth stages of winter wheat. This study concentrated on drought occurring at ten-day intervals during the growth stages. In the study area, rainfed cropland is mainly distributed in the northeastern part of the plains, and irrigated cropland is primarily distributed in the western and southern regions. Precipitation is distributed unevenly in space, and in time, it is mainly concentrated in summer. In spring, as the key growth and development stage of winter wheat, precipitation is low and often accompanied by spring drought, resulting in the limited irrigation area of crop fields not being able to meet the water demand of winter wheat during its growth and development stage. As a consequence, drought has become a major limiting factor affecting grain production in the Guanzhong Plain.

B. Remote Sensing Data

1) Sentinel Series Data: The Sentinel series data used in this study are Sentinel-2 MSI Level-2A data and Sentinel-3 sea–land surface temperature radiometer (SLSTR) level-2 LST products and ocean and land color instrument (OLCI) Level-2 land full resolution (LFR) products downloaded from ESA (https://scihub.copernicus.eu/dhus/#/home) under clear sky conditions in 2021. The main time when drought occurs in the Guanzhong Plain is concentrated from March to May, which corresponds to the main growth stages of winter wheat. Therefore, remote sensing images during this period were selected for the study. Meanwhile, due to the influence of clouds, Table I shows the available images under clear sky conditions. Among them, Sentinel-2 acquired a total of four scenes of images, covering late March, late April, early May, and late May, and the data of Sentinel-3 were acquired in the corresponding ten-day intervals, respectively. After resampling to 20 m, reprojecting to the UTM zone 49N, and stitching, the data obtained by Sentinel-2 were used to retrieve NDVI for fusion studies and to retrieve LAI, the fraction of green vegetation cover (FCOVER), and canopy water content (CWC) for downscaling of LST by using the Sentinel application platform (SNAP). For the acquired

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

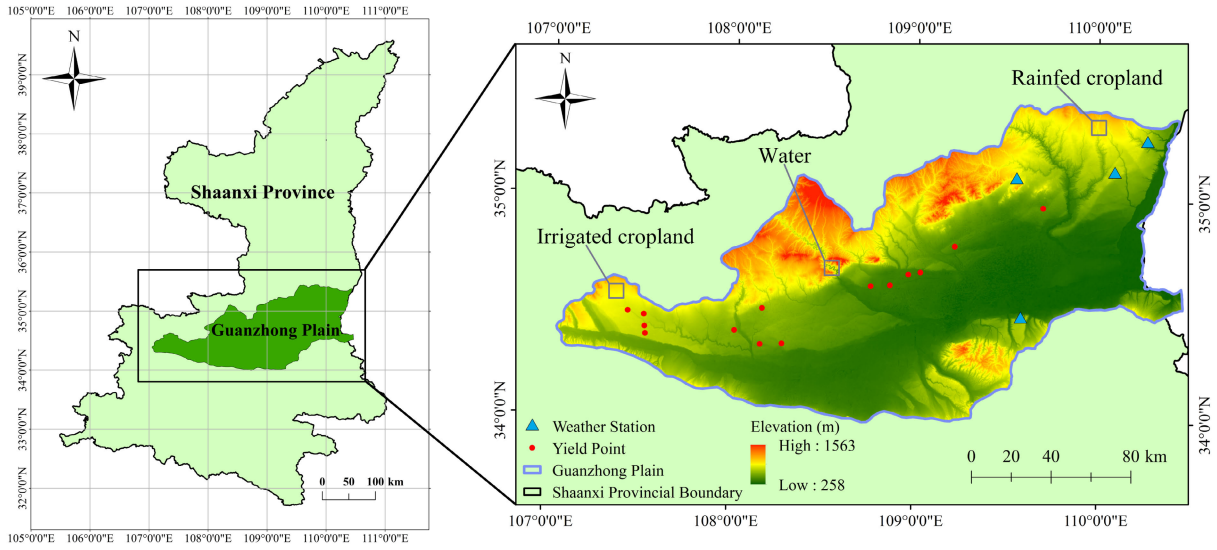


Fig. 1. Location map of the Guanzhong Plain in China, topographic features, and distribution of weather stations, measured sample points for winter wheat cultivation and sample areas.

TABLE I
ACQUISITION DATES OF SENTINEL-2 AND -3 DATA UNDER CLEAR SKY CONDITIONS IN 2021

Time	Sentinel-2	Sentinel-3
Late March	29	23, 27, 28
Late April	28	28, 30
Early May	8	1, 5, 6, 8, 9, 10
Mid-May	—	17, 20
Late May	23	30, 31

Sentinel-3 data, the NDVI was obtained from the LFR data, and the LST was used for downscaling studies, and based on this, Sentinel-3 VTCI was computed, and a fusion study was carried out. Meanwhile, in the LST downscaling process, considering the existence of north-high and south-low topographic features in Guanzhong Plain, the advanced spaceborne thermal emission and reflection radiometer (ASTER) global digital elevation model (DEM) V002 product with a spatial resolution of 30 m (https://search.earthdata.nasa.gov/) was used as an impact parameter for the downscaling study.

2) MODIS Series Data: Taking into account that the available Sentinel series data were acquired during the morning pass, the study also opted for Terra MODIS data captured during the morning transit. The specific data used in this study were the MOD11A1 and MOD09GA data products downloaded from the official NASA website (https://ladsweb.modaps.eosdis.nasa.gov/) for the past 20 years, which were used to calculate the VTCI for multiple years. In this case, the daily LST data from MOD11A1 were selected, the NDVI was calculated using the infrared and near-infrared bands from MOD09GA, and the multiyear warm and cold edges of the VTCIs were determined sequentially using the maximum value composite and minimum value composite, and then, the multiyear ten-day interval VTCIs were calculated.

C. Field Measurement Data

Based on the conclusion that VTCI has a strong correlation with ten-day accumulated precipitation found by Sun et al. [19], in this study, the daily precipitation data from meteorological stations distributed near rainfed cropland in the Guanzhong Plain were selected for the verification and analysis of VTCI drought characterization, and the specific distribution of meteorological stations was shown in Fig. 1. At the same time, in order to further verify the validity of field-scale VTCI in characterizing drought stress in crop growth as reflected in the final yield, the yield data from 14 sample points of the 2021 field survey were selected for verification, and the specific distribution of the sample points was also shown in Fig. 1. The specific steps for measured yield were as follows: first, for each sample point, four persons were assigned, each of whom selected three 60 × 60 cm squares, and a total of 12 squares were collected. The distance between each square was 10 m, and the distance between each person was about 20 m. Second, the total number of spikes of winter wheat in each square was counted, and ten spikes were randomly sampled from each square for sun drying, threshing, and baking at 105 °C for 2 h, followed by baking at 65 °C for 24 h. Finally, winter wheat yields at the sample sites were calculated based on the weight of the dried kernels, the sample square area, and the number of spikes within the sample square [33].

III. METHODS

A. Calculation of Multiyear VTCI

The VTCI is used as a near real-time drought monitoring index on a time scale of ten-day intervals, and it is calculated using the formula shown below [17], [19]

$$\mathrm{VTCI} = \frac{\mathrm{LST}_{\max}(\mathrm{NDVI}_i) - \mathrm{LST}(\mathrm{NDVI}_i)}{\mathrm{LST}_{\max}(\mathrm{NDVI}_i) - \mathrm{LST}_{\min}(\mathrm{NDVI}_i)} \quad (1)$$

$$\mathrm{LST}_{\max}(\mathrm{NDVI}_i) = a + b\mathrm{NDVI}_i \quad (2)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

$$\mathrm{LST}_{\min}(\mathrm{NDVI}_i) = a' + b'\mathrm{NDVI}_i \tag{3}$$

where $\mathrm{LST}_{\max}(\mathrm{NDVI}_i)$ and $\mathrm{LST}_{\min}(\mathrm{NDVI}_i)$ are the maximum and minimum LST values for all pixels in the study area at ten-day intervals when the NDVI value is equal to NDVI_i, which are called the warm and cold edges, respectively. $\mathrm{LST}(\mathrm{NDVI}_i)$ is the LST value when the NDVI value is equal to NDVI_i. The coefficients a, b, a', and b' are approximated by the scatter plots of NDVI and LST in the study area.

The most critical step in the calculation of VTCI is the determination of the warm and cold edges. If only one-year's data are used to determine the warm and cold edges to obtain a single-year's VTCI, it only describes relatively wet and dry conditions and cannot allow for comparisons and analyses of multiyear drought situations. Therefore, Sun et al. [19] proposed a method to determine the warm and cold edges based on multiyear NDVI and LST data successively using the maximum value composite method and the minimum value composite method, and it was proved that the obtained multiyear VTCI was able to characterize the occurrence of droughts as well as interannual drought variations in a near real-time manner. Since the determination of VTCI warm and cold edges requires multiyear data at a ten-day interval, such as MODIS VTCI, which uses the past 22 years of NDVI and LST data in its calculations, and the data vintage period available for the existing Sentinel series data is relatively short, this study determined the warm and cold edges of the multiyear Sentinel-sourced VTCI based on analyzing the relationship between the single-year warm and cold edges of the Sentinel data and the single-year and multiyear warm and cold edges of the MODIS data.

B. Downscaling of LST Based on Random Forest Regression

Random forests, first proposed by Breiman [34], are an ensemble machine learning method based on decision trees. As a modified form of classification and regression tree (CART), the bootstrap method is used to improve model performance. Each tree is built from a random subset of training data and a random subset of predictor variables. Differences between trees are smoothed by averaging the outputs of many trees $T_m(r_i)$ $(m = 1, 2, \ldots, M)$ [see (4)] to increase model stability and reduce sensitivity, as well as to avoid overfitting problems and ultimately produce more accurate predictions $\hat{p}(r_i)$ [35]

$$\hat{p}(r_i) = \frac{1}{M}\sum_{m=1}^{M} T_m(r_i) \tag{4}$$

where r_i denotes the set of i features.

In addition, considering that multiple downscaling features used in this study were derived from the Sentinel-2 data, there was a certain degree of correlation among them. Random forests are highly resistant to interference and can better learn the interactions between multiple features with correlations [36], and the use of random forests for LST downscaling studies has been proven to be well suited [37]. In this study, the biophysical parameters of the Sentinel-2 inversion and the DEM data were first aggregated to 1000 m as training data, and the regression relationship was constructed using RF as follows:

$$\mathrm{LST}_{1000\,\mathrm{m}} = f(r_i) \tag{5}$$

where r_i represented the set of LAI, FCOVER, CWC, and DEM, and f was a nonlinear function fit by the RF algorithm.

Next, the biophysical parameters of the Sentinel-2 and DEM resampled to 20 m were used as inputs to obtain the predicted LST of 20 m ($\widehat{\mathrm{LST}}_{20\,\mathrm{m}}$). On the basis of downscaling using random forest regression, given that it is difficult for the regression model to capture the overall changes in the LST distribution, this study applied the residual correction method proposed by Kustas et al. [38]. The correction procedure consisted of reaggregating the 20-m LST predictions ($\widehat{\mathrm{LST}}_{20\,\mathrm{m}}$) to 1000-m LST ($\widehat{\mathrm{LST}}_{1000\,\mathrm{m}}$) for calculating the LST residuals ($\Delta\mathrm{LST}_{1000\,\mathrm{m}}$), which were resampled and added to the LST ($\widehat{\mathrm{LST}}_{20\,\mathrm{m}}$) to obtain the final downscaled LST values ($\widehat{\mathrm{LST}}_{20\,\mathrm{m}}$)

$$\Delta\mathrm{LST}_{1000\,\mathrm{m}} = \mathrm{LST}_{1000\,\mathrm{m}} - \widehat{\mathrm{LST}}_{1000\,\mathrm{m}} \tag{6}$$

$$\widehat{\mathrm{LST}}_{20\,\mathrm{m}} = \widehat{\mathrm{LST}}_{20\,\mathrm{m}} + \Delta\mathrm{LST}_{1000\,\mathrm{m}}. \tag{7}$$

The specific data usage was that data obtained by Sentinel-2 on March 29, April 28, May 8, and May 23 and DEM were used as inputs, and the corresponding Sentinel-3 LSTs of March 28, April 28, May 8, and May 20 were used as labels to achieve downscaling studies of the LSTs.

C. STARFM

STARFM is a fusion method for generating comprehensive high-resolution images proposed by Gao et al. [25], and its basic assumption is that the systematic deviation between two sensors does not vary with time. This method uses a pair of high-/low-resolution images at the reference time and low-resolution images at the target time for fusion. Specifically, in the fusion process, the influence of the pixel boundary of the low spatial resolution image is reduced by introducing a moving window and similar pixels, and the value of the center pixel of the window is calculated by considering the spatial weight, spectral weight, and temporal weight together using the similar pixels in the window. It can be expressed by the following formula:

$$\begin{aligned} &F\left(x_{\frac{w}{2}}, y_{\frac{w}{2}}, t_0\right) \\ &= \sum_{i=1}^{w}\sum_{j=1}^{w}\sum_{k=1}^{n} W_{ijk} \\ &\quad \times \left(C\left(x_i, y_j, t_0\right) + F\left(x_i, y_j, t_k\right) - C\left(x_i, y_j, t_k\right)\right) \end{aligned} \tag{8}$$

where x_i and y_j are the given pixel location, and F and C represent high spatial resolution low temporal resolution images and high temporal resolution low spatial resolution images, respectively. Also, w is the searching window size, and $(x_{(w/2)}, y_{(w/2)})$ is the central pixel of this moving window. t_0 and t_k stand for the prediction date and the reference image date, respectively, where $k \in [1, n]$. It is worth noting that W_{ijk} is a combined weighting function based on spectral, spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

distance, and time information and is as follows:

$$W_{ijk} = \frac{\left(\frac{1}{(S_{ijk} * D_{ijk} * T_{ijk})}\right)}{\sum_{i=1}^{w}\sum_{j=1}^{w}\sum_{k=1}^{n}\left(\frac{1}{S_{ijk} * D_{ijk} * T_{ijk}}\right)} \tag{9}$$

where S, D, and T indicate the spectral similarity, spatial distance, and time difference between the fine and coarse images, respectively.

In this study, Sentinel-2 data (20 m) were used as high spatial resolution data, and Sentinel-3 data (1000 m) were used as low spatial resolution data. According to the land use situations in the Guanzhong Plain during the study time and the existing research experience of scholars, the width of the moving window (w) was set to 101 pixels of the high spatial resolution image, which is equivalent to two pixels of the low spatial resolution image, by trying different window sizes and comparing the fusion results.

D. Accuracy Assessment

Fusion or downscaling image quality evaluation methods are mainly categorized into two types: a subjective evaluation method that uses human eye vision as the main evaluation index and an objective evaluation method that uses specific algorithms to provide specific quantitative indices. Therefore, three commonly used objective quantitative evaluation metrics were selected in this study to evaluate the results of fusion and downscaling. The root-mean-square error (RMSE) is used to calculate the difference between expected and actual values and to describe the overall prediction error. The structural similarity index (SSIM) is applied to compare the overall structural similarity of the original and fused images. The correlation coefficient (r) is employed to demonstrate the linear relationship between expected and observed data. Theoretically, the smaller the RMSE value and the larger the SSIM and r values, the more accurate the fusion result is. It should be noted that since neither the LST nor the ten-day interval VTCI has corresponding real data at high spatial resolution, the obtained data at high spatial resolution need to be resampled to coarse spatial resolution when the metrics are utilized to validate their downscaling or fusion accuracies, and to verify their validity and consistency.

IV. Results and Analysis

A. Calculation of Multiyear Sentinel-3 VTCI

1) Determination of the Warm and Cold Edges of the Sentinel-3 VTCI: The warm and cold edges of the VTCI are mainly composed of slope and intercept. In this case, the slope indicates the extent to which the change in the unit independent variable affects the dependent variable, i.e., the degree to which the change in NDVI affects the LST. The intercept indicates the base level of the dependent variable when the independent variable is zero, i.e., the predicted baseline value of LST when NDVI is zero. Therefore, this study compared the corresponding slope and intercept changes of the warm and cold edges of the 2021 Sentinel-3 VTCI, the 2021 MODIS VTCI, and the multiyear MODIS VTCI under the available data conditions, which, in turn, determined

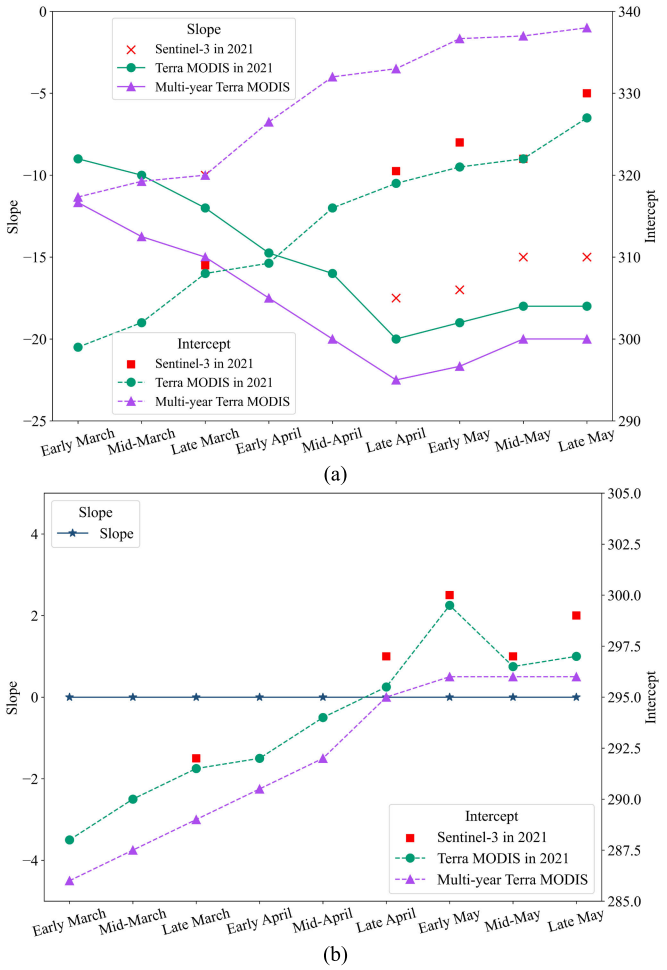


Fig. 2. Variation of the intercepts and slopes of (a) warm edges and (b) cold edges for Terra MODIS and Sentinel-3 VTCI.

the warm and cold edges of the Sentinel-3 multiyear VTCI, as shown in Fig. 2. By analyzing the changes in the slope and intercept of the warm edge [see Fig. 2(a)], it can be seen that the slopes and intercepts of the warm edge determined by single-year MODIS and single-year Sentinel-3 data in 2021 had the same trend of change, and took similar values. It demonstrated that single-year NDVI had a similar influence on LST under different data sources, and LST had a similar prediction basis when NDVI was 0. Meanwhile, comparing the slopes of the warm edges determined from single-year and multiyear MODIS data, it was found that the overall trend was first decreasing and then increasing, with a more obvious decrease from late March to late April and reaching the lowest point in late April. Also, the intercepts showed an overall increasing trend. However, further analysis revealed that based on the changes having consistency, the slope and intercept changes of the warm edge determined using multiyear MODIS data were more stable than those determined using single-year MODIS data, which could overcome the uncertainty caused by the changes of single-year data. Thus, in combination with the above analysis, it is feasible to calculate the multiyear Sentinel-3 VTCI using the warm edge determined by multiyear MODIS data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CONSISTENCY BETWEEN MULTIYEAR SENTINEL-3 AND TERRA MODIS VTCI CALCULATED USING MULTIYEAR MODIS WARM AND COLD EDGES

Time	RMSE	r	SSIM
Late March	0.06	0.79	0.59
Late April	0.14	0.70	0.46
Early May	0.08	0.77	0.55
Mid-May	0.10	0.61	0.45
Late May	0.10	0.74	0.45

Fig. 2(b) illustrates the variability of the cold edge determined by using Sentinel-3 and MODIS data for the single year of 2021 as well as by utilizing multiyear MODIS data. In this case, since the cold edge indicates that the soil moisture is sufficient and the crop growth is no longer affected by water stress, i.e., no matter how the NDVI changes, its corresponding LST will not change and keep a constant value. Therefore, the slope of the cold edge is 0, and this study would primarily focus on the variation of the cold edge intercept. From the figure, it can be seen that the cold-edge intercepts determined using single-year and multiyear MODIS show a gradual increase over time. Among them, the cold-edge intercept of single-year MODIS in early May has a significant increase compared with the smoother changing intercept of multiyear MODIS, while the cool edge of single-year Sentinel-3 in early May has more consistency with the change of single-year MODIS VTCI. Therefore, based on the consistency of the overall trend and the similarity of the cold edge intercepts determined from the single year of data in early May, it further demonstrated that the cold edge determined from the multiyear MODIS data can be used as the cold edge for the Sentinel-3 multiyear VTCIs, resulting in Sentinel-3 VTCIs that are comparable from year to year.

2) Quantitative Evaluation of Multiyear Sentinel-3 VTCI: Multiyear Sentinel-3 VTCIs were obtained using the warm and cold edges of the multiyear MODIS VTCIs and further evaluated for consistency between the multiyear MODIS and Sentinel-3 VTCIs, and the results were presented in Table II.

As depicted in Table II, the RMSE between the multiyear Sentinel-3 VTCI and the multiyear MODIS VTCI ranged from 0.06 to 0.14, r ranged between 0.61 and 0.79, and the SSIM consistently exceeded 0.45, which showed a high degree of agreement. It was noteworthy that in late April and mid-May, the deviations between Sentinel-3 and MODIS VTCIs were large, with RMSE greater than 0.10, r less than 0.70, and SSIM less than 0.50. The reason for this is that, on the one hand, there is a large difference in the slopes of the warm edges of single-year Sentinel-3 and multiyear MODIS in late April and mid-May compared with other times, and a large gap between the intercepts of the warm edges in mid-May, which, in turn, leads to the fact that Sentinel-3 VTCIs directly derived from the multiyear warm-edge of MODIS cannot adequately reflect the influence of NDVI on the LSTs, and, thus, affects the consistency of the Sentinel-3 VTCI with the MODIS VTCI in late April and mid-May. On the other hand, there is a systematic bias in the LST data obtained due to the different transit times of the two sensors, which also created a bias in the direct application of the multiyear MODIS cold edges to the multiyear Sentinel-3 VTCI calculations. The combination of these two aspects resulted in less consistent VTCI results for late April and mid-May.

B. Downscaling of LST

In validating the accuracy of the LST after downscaling, it is considered that the real acquired LST data of 20 m do not exist yet. Therefore, in this study, the 20-m LST after downscaling was aggregated to 1000 m and compared with the directly acquired Sentinel-3 LST to plot the scatter density maps, and the corresponding results under each acquisition date are shown in Fig. 3. As can be seen in Fig. 3, the scatterplot distributions of the downscaled LST data and the Sentinel-3 LST data at different dates were approximately around the 1:1 line. Meanwhile, at different dates, it can be clearly seen that the data points were clustered in the center region of the plot, forming an obvious high-density region, and its existence indicated that there was a close positive correlation between the downscaled LST and the Sentinel-3 LST. However, it was worth noting that the slopes of the fitted lines were usually smaller than the 1:1 line, which suggested that the downscale LST data had obvious underestimation for the Sentinel-3 LST. The reason for this phenomenon is that, on the one hand, in the construction of the random forest downscaling model, it is necessary to aggregate the high spatial resolution feature variables to 1000 m using pixel aggregation, which leads to the underestimation by neglecting the detailed features indicating changes in LST. On the other hand, due to the small distribution of the number of LST data in the high-value interval, the model cannot fully learn the correspondence in the learning process, which led to the underestimation phenomenon for high-value LST. Further comparison of the downscaled accuracies of the LSTs for each date revealed that the r values of the downscaled LSTs and the LSTs of the Sentinel-3 were above 0.70, the RMSEs were all less than 1.80 K, and the SSIMs were all up to 0.98 for different dates, which showed a high degree of consistency. It was noted that on April 28 [see Fig. 3(b)], the accuracy of the downscaled LST was relatively low, and the scatter was relatively sparse compared with other dates, which was since the feature data acquired on April 28 were more affected by clouds than other dates, resulting in a relatively small amount of data, which affected the accuracy on this day.

C. Spatiotemporal Data Fusion for VTCI

Since the calculation of VTCI involves both NDVI and LST, there can be two different fusion schemes to achieve the fusion study of VTCI. The specific fusion schemes are shown in Fig. 4. Scheme 1: based on the BI strategy, the NDVI of Sentinel-3 1000 m and Sentinel-2 20 m, and the Sentinel-3 1000-m LST and the LST downscaled to 20 m were fused using STARFM, respectively, to obtain the fused NDVI and LST at ten-day intervals for 20 m. On this basis,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

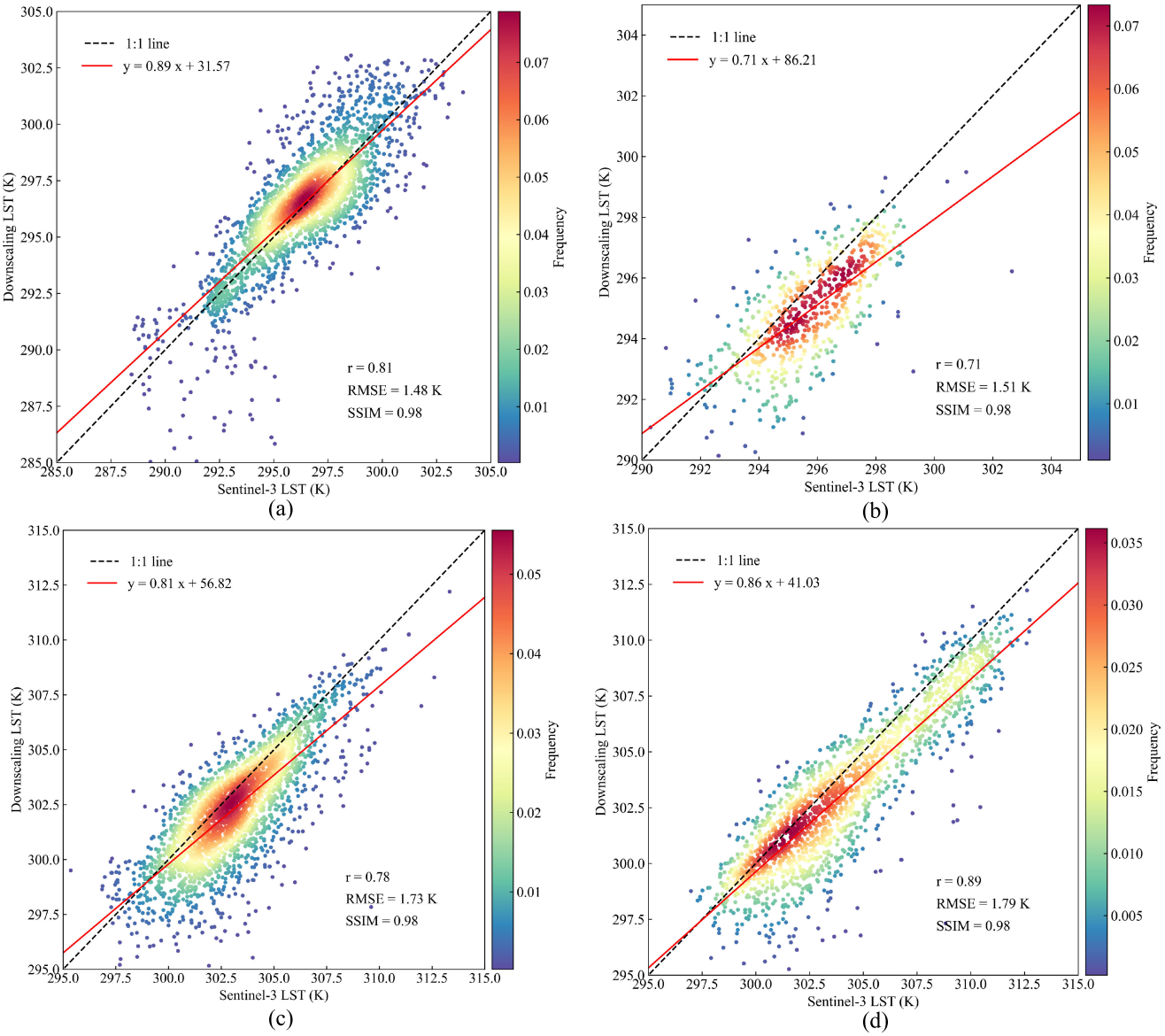


Fig. 3. Scatter density plots of Sentinel-3 LST and downscaled LST on (a) March 29, (b) April 28, (c) May 8, and (d) May 23.

the VTCI was obtained at ten-day intervals for 20 m after fusion according to the formula for VTCI. Scheme 2: based on the IB strategy, on the premise of using NDVI and LST to calculate 1000-m VTCI for ten days and 20-m VTCI for one day, the STARFM algorithm was used to directly fuse the VTCI to obtain 20-m VTCI for ten days. After obtaining VTCI with 20-m spatial resolution at ten-day intervals through the two fusion schemes mentioned above, this study validated the effectiveness of the field-scale VTCI based on the fusion of the different schemes by using the ten-day cumulative precipitation at the weather stations corresponding to the rainfed sample sites and the measured yields at the sample site scale.

1) Accuracy Validation of Fused NDVI and LST in Scheme 1: Since the fusion of NDVI and LST is required before fusion using Scheme 1, the accuracies of the fused NDVI and LST were first evaluated in this study. Since the fused NDVI and LST were 20 m at ten-day intervals, which lacks validation data with high spatial and temporal resolution, the fused NDVI and LST were further aggregated to 1000 m and compared with the Sentinel-3 NDVI and LST at ten-day intervals to validate the fusion accuracy, and the results are shown in Table III. The analysis indicated that the RMSE of NDVI for every ten days before and after fusion ranged from 0.07 to 0.14, the r value ranged from 0.34 to 0.77, and the SSIM was greater than 0.75, which was a high degree of consistency. Meanwhile, it should be noted that the relatively low accurate results in late March and late April were due to the relatively small amount of NDVI data available in that time interval, while the number of pixels was further reduced by removing cloudy pixels, thus leading to the low accuracy of the fusion results. For the LST, after the fusion using STARFM, it was found that, except for late May, the LSTs after fusion and the original Sentinel-3 ten-day LSTs were able to maintain good agreement, in which the RMSEs were all less than 1.50 K, the r values were all above 0.85,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

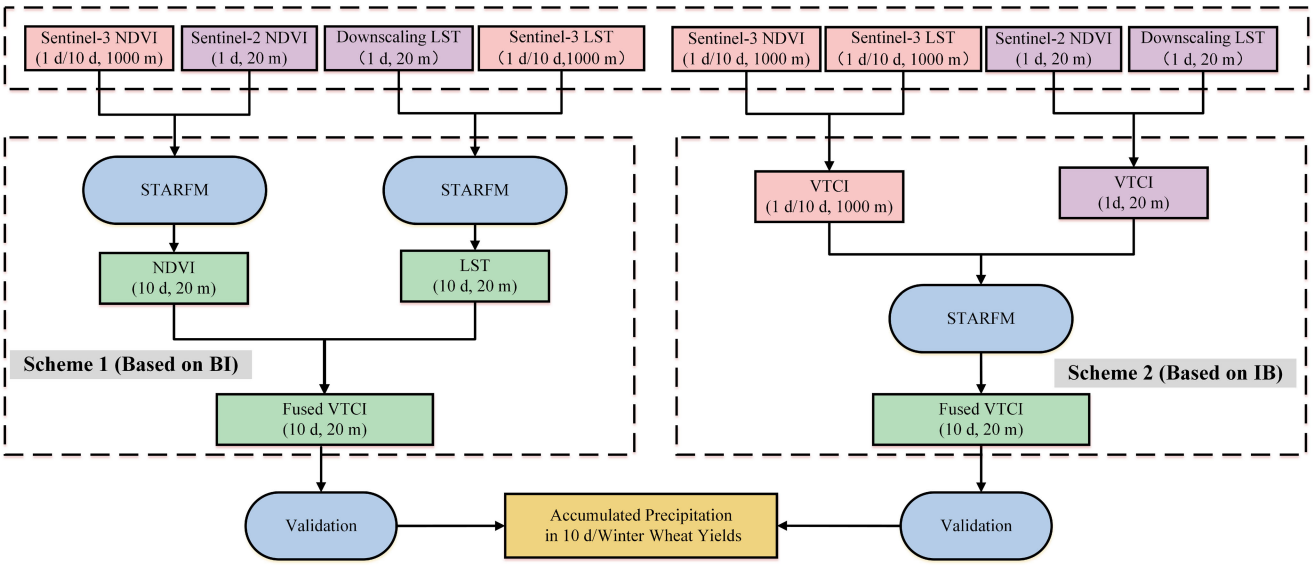


Fig. 4. Schematic of the fusion schemes.

TABLE III
ACCURACY VALIDATION OF FUSED NDVI AND LST

Time	NDVI			LST		
	RMSE	r	SSIM	RMSE (K)	r	SSIM
Late March	0.14	0.34	0.88	1.22	0.85	0.74
Late April	0.11	0.49	0.75	1.46	0.85	0.79
Early May	0.08	0.61	0.90	1.08	0.94	0.82
Mid-May	0.11	0.42	0.90	1.32	0.93	0.80
Late May	0.07	0.77	0.97	3.72	0.88	0.69

and SSIM was all greater than 0.74. The reason for the large error in the LST before and after fusion in late May, on the one hand, was limited by the quality of the acquired data. On the other hand, the data from May 23 and May 30 were utilized as reference images in the fusion to achieve the fusion for LST at ten-day intervals. The long-time interval between the reference image pairs further affected the fusion accuracy at this time. The above analysis proved that both NDVI and LST after fusion were in better agreement with the NDVI and LST of the Sentinel-3 ten-day interval, which can be used to calculate the VTCI.

2) Comparison of the Two Fusion Schemes: A total of five sets of VTCI fusion experiments at ten-day intervals were conducted with the 2021 data set. The results before and after the fusion of VTCI in the entire Guanzhong Plain are shown here as an example in early May (see Fig. 5). It can be seen that the distributions of VTCI based on the fusion of the two schemes were identical to that of Sentinel-3 throughout the Guanzhong Plain. For example, in the northeastern parts of the Plain, the VTCI values were relatively small, all lower than 0.4, and there was a more obvious drought phenomenon. The main reason for this phenomenon was that the northeastern region was dominated by rainfed croplands and heavily dependent on precipitation. However, precipitation was relatively low in this region, thus leading to pronounced drought problems. In marked contrast to the above phenomenon was in the western part of the Guanzhong Plain. As the western region was relatively rich in water resources due to the flow of the Wei River, the VTCI values were relatively high, which made the drought conditions relatively mild.

In order to further compare the fusion results of the two fusion schemes, representative irrigated cropland, rainfed cropland, and water, with specific location distribution shown in Fig. 1, were selected for display, as shown in Fig. 6. Through observation, it can be seen that in the selected representative areas, compared with the VTCI of Sentinel-3, the VTCIs obtained based on the two fusion schemes can show the distribution of land use types more clearly, which proves the effectiveness of the fusion. Among them, in the irrigated cropland area [see Fig. 6(a)], the fused VTCI based on Scheme 1 can clearly demonstrate the boundaries between fields and the differences between the VTCI values of different fields. In contrast, the boundary differentiation between VTCI fields obtained using Scheme 2 was relatively low, and the display of differences between VTCI values was relatively fuzzy. In the rainfed cropland area [see Fig. 6(b)], compared with the fused VTCI obtained based on Scheme 2, the fused VTCIs based on Scheme 1 were able to characterize the drought differences more clearly under different rainfed fields and showed more detailed texture details. For the zigzag water [see Fig. 6(c)], the fused images were able to show the flow characteristics of the water body more clearly. In the fused

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

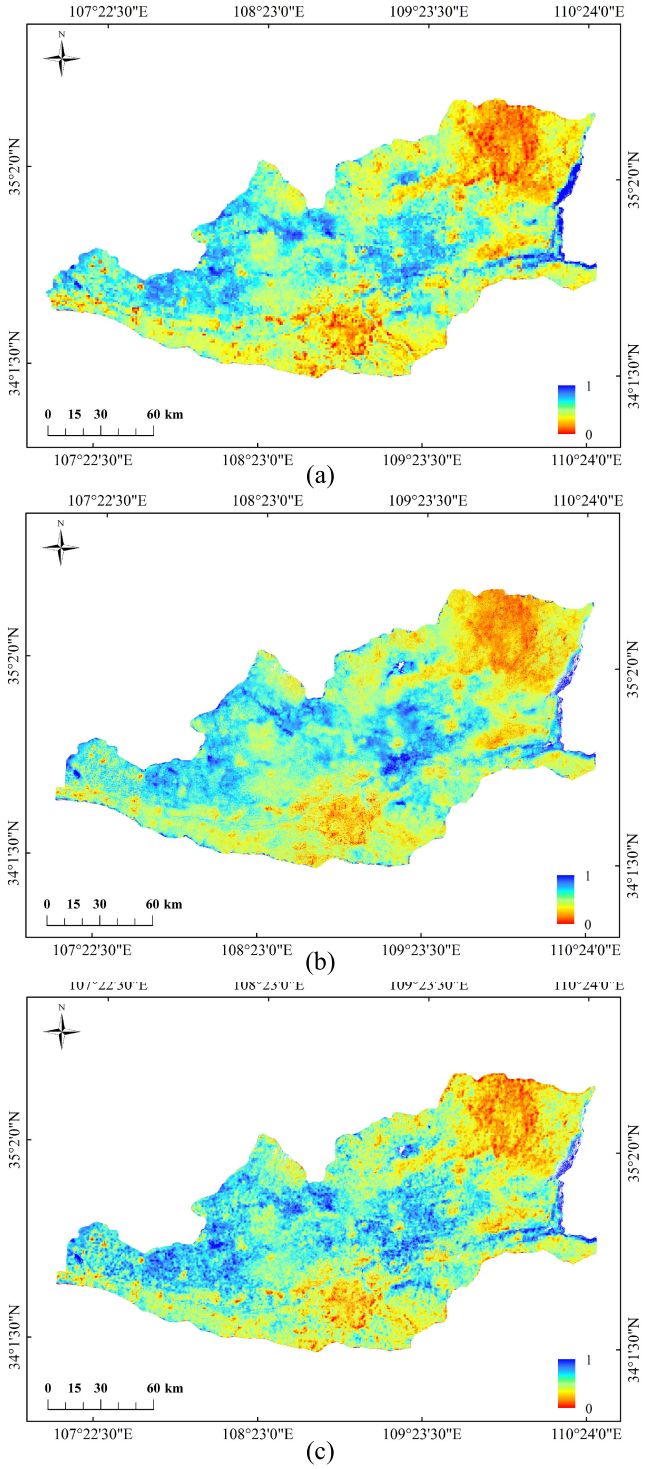


Fig. 5. Comparison of early May VTCI before and after fusion based on two schemes. (a) Sentinel-3 VTCI. (b) VTCI obtained based on Scheme 1. (c) VTCI obtained based on Scheme 2.

results of Scheme 2, the display boundary of the zigzag water body portion was still blurred. However, the fused VTCI generated according to Scheme 1 has a more complete and clearer boundary of the water body, as well as a better reflection of the drought conditions in the area surrounding the water. The above qualitative comparative analysis of different fusion schemes for fusing VTCIs in specific regions indicated that compared with fused VTCIs based on Scheme 2, fused VTCIs based on Scheme 1 had stronger regional differentiation as well as the ability to characterize drought differences.

TABLE IV
COMPARISON OF THE CONSISTENCY OF THE DIFFERENT SCHEMES FUSING VTCI WITH SENTINEL-3 VTCI

Time	Scheme	RMSE	r	SSIM
Late March	1	0.05	0.86	0.68
	2	0.05	0.88	0.66
Late April	1	0.06	0.85	0.69
	2	0.05	0.84	0.63
Early May	1	0.04	0.93	0.79
	2	0.04	0.88	0.67
Mid-May	1	0.04	0.95	0.82
	2	0.05	0.91	0.71
Late May	1	0.12	0.84	0.65
	2	0.14	0.82	0.57

Based on the above qualitative analysis, the fusion accuracy was further evaluated from a quantitative perspective. Due to a lack of corresponding high spatial and temporal resolution VTCIs for comparison, the fused VTCIs were again aggregated to the 1000-m spatial resolution and validated for consistency with the VTCIs of Sentinel-3 at the ten-day intervals. The results are shown in Table IV. From the table, it showed that the RMSE values were between 0.04 and 0.14, with r being greater than 0.82 and SSIM being greater than 0.57, i.e., the fusion VTCIs obtained using the two schemes showed good concordance in each of the ten-day intervals. In a specific comparison of the fusion results corresponding to each ten-day fusion scheme, it was found that the values of RMSE in concordance between the VTCI obtained based on Scheme 1 and the Sentinel-3 VTCI were lower than those calculated based on Scheme 2, except in late April. The correlations between the VTCI obtained by both schemes and the original VTCI were more similar. Comparing from the perspective of SSIM, it can be found that the VTCI obtained based on Scheme 1 had higher spatial similarity with the Sentinel-3 VTCI. This was because Scheme 2 did not take into account the changes in subpixel landscapes over time, resulting in a large error in the fusion results. The direct fusion of high spatial resolution VTCIs cannot handle abrupt changes in different land cover types, thus increasing uncertainty. In contrast, in Scheme 1, the high spatial resolution VTCI was calculated directly from the fused NDVI and LST, which can reduce the effects of localized clouds and aerosol contamination, thus improving data quality and overall performance. The strategy of calculating NDVI and LST before fusion can effectively reduce noise and outliers. The results of this study were in accordance with the findings of previous scholars [32], [39], [40].

3) Validation of Drought Monitoring Results Using the Fused VTCI: VTCI, as a near real-time drought monitoring index, can respond to the water stress suffered by the crop during the growth process, and Sun et al. [19] showed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

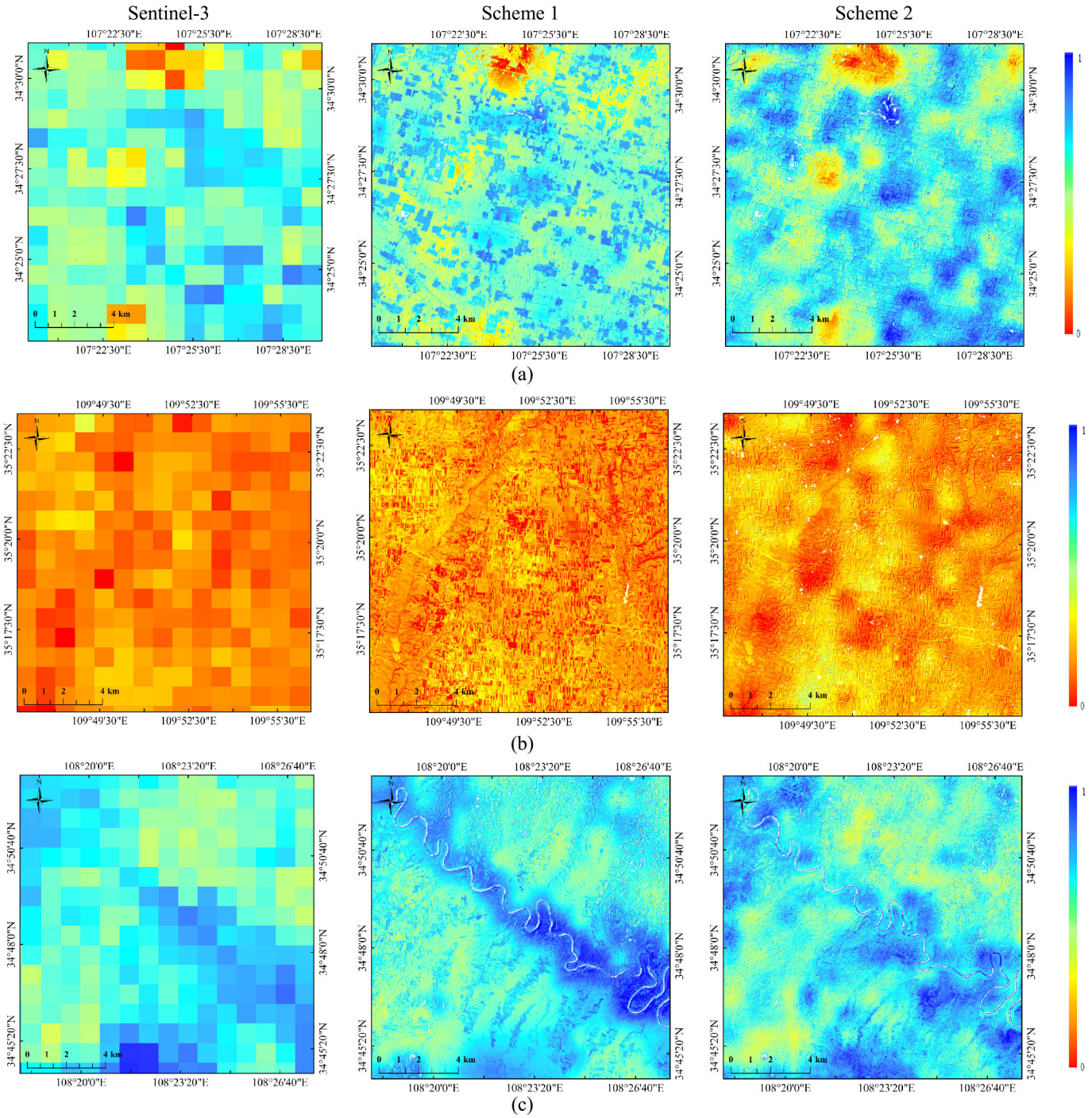


Fig. 6. Spatial comparison of the VTCIs estimated by the two fusion schemes with the corresponding Sentinel-3 VTCIs for different land use types in early May, which are (a) irrigated cropland, (b) rainfed cropland, and (c) water.

that the ten-day VTCI had a strong correlation with the ten-day cumulative precipitation. Therefore, to further verify the degree of response of VTCI to precipitation after fusion, this study calculated the correlation between ten-day cumulative precipitation and VTCI at weather stations distributed in the Guanzhong Plain adjacent to the rainfed cropland sites (see Fig. 1 for the specific distribution), and the results are shown in Fig. 7. The analysis showed that the VTCI with a spatial resolution of 1000 m in Sentinel-3 had the highest correlation with ten-day cumulative precipitation, which was able to reach 0.51. The correlations with precipitation based on the VTCI were also higher and similar after the fusion of schemes 1 and 2, at 0.30 and 0.44, respectively. Namely, the sensitivity of the field-scale VTCI response to precipitation was lower than that of the VTCI before fusion. This phenomenon may be because the data obtained from the weather stations can express the precipitation changes at the kilometer scale of the stations, and the spatial resolution of the VTCI of the Sentinel-3 is 1000 m, so the two scales of characterization are similar. However, the fused VTCI, with a resolution of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

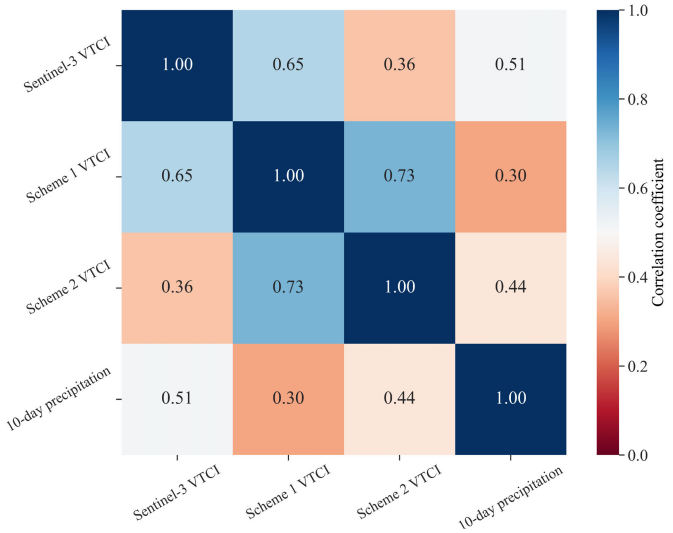


Fig. 7. Correlation of VTCI before and after fusion with ten-day cumulative precipitation.

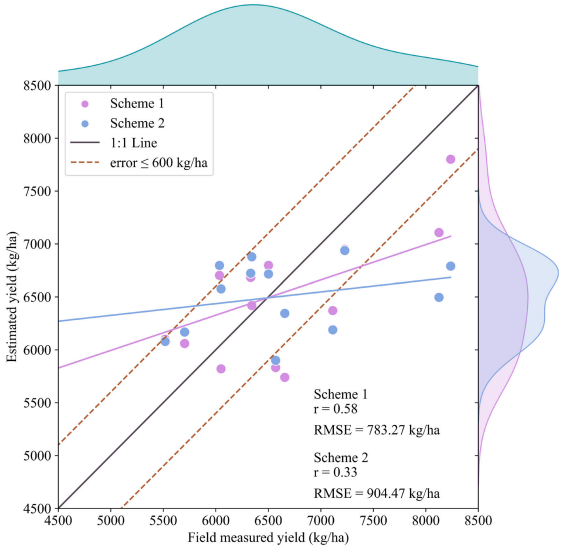


Fig. 8. Sample point scale yield estimation results based on fused VTCI.

20 m, is more focused on characterizing moisture changes at the field scale, and there is a spatial-scale mismatch with the accumulated precipitation data obtained from the stations.

In recent years, based on drought monitoring using VTCI, it has been further applied to yield estimation research and achieved high accuracy in estimating yield [5]. Also, to further compare the applicability of the two fusion schemes, in this study, the fused VTCI was utilized to estimate the yield of the sample sites measured in the field of the Guanzhong Plain in 2021, to verify the applicability of different fusion schemes for the yield estimation of the sample sites. Due to the relatively small number of sample points, the simplest linear regression model was used for yield estimation in this study, and the specific results were shown in Fig. 8. As can be seen from Fig. 8, it showed that the estimation model constructed using the fused VTCI of Scheme 1 had an r of 0.58 and an RMSE of 783.27 kg/ha. The estimation model constructed using the fused VTCI of Scheme 2 had an r of 0.33 and an RMSE of 904.47 kg/ha. The accuracy of the estimation model constructed using Scheme 1 was higher. By comparing the distribution plots of the estimated yields realized by VTCI based on different fusion schemes and measured yields, it was found that the distribution of the estimated yields based on Scheme 1 was more similar to the distribution plots of the measured yields, which were all normally distributed. At the same time, compared with Scheme 2, the line of fit between measured yields and estimated yields of Scheme 1 was closer to the 1:1 line, and more points fell within the margin of error of 600 kg/ha. Moreover, compared with Scheme 1, there was an obvious underestimation of high yields and an overestimation of low yields in the results obtained based on Scheme 2. This is further evidence of the higher accuracy of the VTCI obtained using Scheme 1 and its good applicability for estimating yields.

V. Discussion

This study proposed a technical framework for field-scale VTCI fusion based on Sentinel series data, from the generation of Sentinel series VTCIs to the investigation of index-based VTCI fusion based on different fusion schemes using the STARFM, and finally, the validity of the fusion was verified by using the ten-day cumulative precipitation data from the meteorological stations as well as yield estimation of the actual winter wheat planting sample sites. Although numerous studies have been conducted to explore the applicability of different fusion schemes, relatively few studies have been conducted on index-based index fusion, and even fewer on VTCI fusion in particular. At the same time, in the process of conducting this study, the limitations of this study as well as targeted improvement measures were further identified as follows.

1) In the calculation process of the Sentinel series VTCI, since the VTCI needs to utilize the NDVI and LST data of the past multiple years for the determination of the warm and cold edges, however, due to the limitation of the existing data volume of the Sentinel, this study analyzed and compared the relationship between the single-year Sentinel and the changes of the warm and cold edges of the MODIS over a single year and multiple years, and then, the multiyear warm and cold edges of the Terra MODIS were directly attributed to the calculation of the multiyear Sentinel VTCI. Although the multiyear Sentinel VTCI was in good agreement with the MODIS VTCI, it still exhibited a large error at the time when the changes of NDVI and LST were more significant. Therefore, in the next step, with the accumulation of the Sentinel series data, we can try to utilize the multiyear data for the determination of the warm and cold edges of the Sentinel series VTCI, so as to reduce the errors and uncertainties brought by data sources at the level of VTCI calculation.
2) In VTCI fusion, it is necessary to utilize LST data with high spatial resolution, and the spatial resolution of LST is lower than that of the visible band due to the limitation of sensor technology, so it is necessary to downscale LST to obtain LST data of 20 m. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

study was centered on the Sentinel series data, so it was necessary to use Sentinel-2 retrieved performance biophysical parameters and DEM data to characterize the terrain. Although the consistency between the downscaled LST and the original LST was high, there was still room for improvement in the accuracy compared with other downscaled fusion models, such as those proposed using multiparameter or specifically for LST [36]. Therefore, in order to further reduce the errors due to LST downscaling, it is possible to try to utilize multiple parameters as well as a targeted fusion to realize the downscaling study of LST.

3) In the course of the study, we attempted to use a variety of fusion methods for VTCI fusion research, such as ESTARFM, FSDAF, and improved FSDAF. However, in practice, we found that since the study area is the Guanzhong Plain with an area of 3.6×10^4 km^2, the fusion image range is large, and the starting point of this study is to explore the applicability of fused VTCIs in the entire Guanzhong Plain. Therefore, we chose the classical STARFM using only one pair of reference images for the VTCI fusion study, based on the comprehensive consideration of the availability of the data, the time cost of the fusion process, and the land types of the Guanzhong Plain at the time of the occurrence of the drought. Although the consistency of the VTCI before and after fusion is relatively high, the STARFM method itself is limited by the fact that it only considers the linear change relationship, which also affects the final fusion results. Therefore, based on this study to verify the effectiveness of fused VTCI in the Guanzhong Plain, the applicability of different methods can be explored in the future by selecting specific regions in a targeted manner.

Overall, it was found that compared with the direct fusion of VTCI, fusing NDVI and LST before calculating VTCI could more clearly characterize the texture information of various types of features and realize the estimation of the yields of the sample points with high accuracy. The results of this study are consistent with the results of Ma et al. [39] and Bei et al. [40] for the fusion of terrestrial latent heat flux, which further proves the applicability of this fusion scheme for the fusion of index-based indices. By fusing the base indices (NDVI and LST) after computation, the noise contained in the base information, such as reflectance, can be attenuated during the computation process, which reduces the error in the fusion process [31]. In contrast, when fusion is performed directly for the VTCI, it may result in the information that changes over time being filtered out, which leads to the limited accuracy of the fusion results. This also proves that the applicability of different fusion schemes needs to be considered when fusing indices based on multi-index computation. Therefore, in the next study, the applicability of the fusion scheme needs to be further verified specifically for different fusion methods.

VI. Conclusion

To move forward understanding of the important of monitoring drought at the field scale, this study proposes a framework for the development of a near real-time drought monitoring approach by using spatiotemporal data fusion to obtain the index-based VTCI of 20 m at ten-day intervals in a large agricultural plain in China (Guanzhong Plain). The following conclusions were drawn from the study.

Based on analyzing the consistency between the single-year Sentinel-3VTCI and the changes in the warm and cold edges of the single-year and multiyear MODIS VTCI, the warm and cold edges of the MODIS multiyear VTCI were directly used in the calculation of the multiyear Sentinel-3 VTCI. The acquired Sentinel-3 VTCI has high consistency and correlation with the MODIS VTCI and ten-day cumulative precipitation and further verified the validity of the methodology of this study. Meanwhile, the acquired VTCI from the Sentinel data source further adds to the existing VTCI remote sensing data sources.

All of the fused VTCIs contained more detailed information and maintained a higher correlation with the ten-day precipitation values. Furthermore, when comparing the applicability of different fusion schemes, compared with directly utilizing VTCIs with different resolutions to achieve fusion, the fusion results obtained by directly performing VTCI calculations after fusion of NDVI and LST are more accurate, more consistent with the original Sentinel-3 VTCIs, and more accurate in crop yield estimation at the sample point scale. The above research also lays the foundation for the next step to continue to explore in depth the applicability of different fusion methods for VTCI, and with potential to design fusion algorithms that are specifically applicable to VTCI, so as to improve the accuracy of fused VTCI and to realize accurate field-scale drought monitoring.

References

[1] United Nations Office for Disaster Risk Reduction. (2021). *GAR Special Report on Drought 2021*. [Online]. Available: https://www.undrr.org/gar2021-drought

[2] United Nations Convention to Combat Desertification. (2022). *Drought in Numbers 2022—Restoration for Readiness and Resilience*. [Online]. Available: https://www.unccd.int/resources/publications/drought-numbers

[3] N. Alahacoon and M. Edirisinghe, "A comprehensive assessment of remote sensing and traditional based drought monitoring indices at global and regional scale," *Geomatics, Natural Hazards Risk*, vol. 13, no. 1, pp. 762–799, Dec. 2022, doi: 10.1080/19475705.2022.2044394.

[4] N. E. Korres, J. K. Norsworthy, N. R. Burgos, and D. M. Oosterhuis, "Temperature and drought impacts on Rice production: An agronomic perspective regarding short- and long-term adaptation measures," *Water Resour. Rural Develop.*, vol. 9, pp. 12–27, Jun. 2017, doi: 10.1016/j.wrr.2016.10.001.

[5] Y. Xie et al., "Assimilation of the leaf area index and vegetation temperature condition index for winter wheat yield estimation using Landsat imagery and the CERES-wheat model," *Agricult. Forest Meteorol.*, vol. 246, pp. 194–206, Nov. 2017, doi: 10.1016/j.agrformet.2017.06.015.

[6] A. Dai, "Drought under global warming: A review," *WIREs Climate Change*, vol. 2, no. 1, pp. 45–65, Jan. 2011, doi: 10.1002/wcc.81.

[7] X. Liu, X. Zhu, Y. Pan, S. Li, Y. Liu, and Y. Ma, "Agricultural drought monitoring: Progress, challenges, and prospects," *J. Geograph. Sci.*, vol. 26, no. 6, pp. 750–767, Jun. 2016, doi: 10.1007/s11442-016-1297-9.

[8] W. Palmer, "Meteorological drought," Weather Bureau, Washington, DC, USA, Res. Paper 45, Jan. 2006, pp. 1–65.

[9] T. B. McKee, N. J. Doesken, and J. R. Kleist, "The relationship of drought frequency and duration to time scales," Presented at the 8th Conf. Appl. Climatol., Anaheim, CA, USA, 1993.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.

[10] H. Wu, M. J. Hayes, A. Weiss, and Q. Hu, "An evaluation of the standardized precipitation index, the China-Z index and the statistical Z-score," *Int. J. Climatol.*, vol. 21, no. 6, pp. 745–758, May 2001, doi: 10.1002/joc.658.

[11] S. M. Vicente-Serrano, S. Beguería, and J. I. López-Moreno, "A multiscalar drought index sensitive to global warming: The standardized precipitation evapotranspiration index," *J. Climate*, vol. 23, no. 7, pp. 1696–1718, Apr. 2010, doi: 10.1175/2009jcli2909.1.

[12] G. Bodner, A. Nakhforoosh, and H.-P. Kaul, "Management of crop water under drought: A review," *Agronomy Sustain. Develop.*, vol. 35, no. 2, pp. 401–442, Apr. 2015, doi: 10.1007/s13593-015-0283-4.

[13] A. AghaKouchak et al., "Remote sensing of drought: Progress, challenges and opportunities," *Rev. Geophys.*, vol. 53, no. 2, pp. 452–480, Jun. 2015, doi: 10.1002/2014rg000456.

[14] P. Shine, T. Scully, J. Upton, and M. D. Murphy, "Multiple linear regression modelling of on-farm direct water and electricity consumption on pasture based dairy farms," *Comput. Electron. Agricult.*, vol. 148, pp. 337–346, May 2018, doi: 10.1016/j.compag.2018.02.020.

[15] A. K. Chaturvedi, U. Surendran, G. Gopinath, K. M. Chandran, A. Nk, and M. F. Ct, "Elucidation of stage specific physiological sensitivity of okra to drought stress through leaf gas exchange, spectral indices, growth and yield parameters," *Agricult. Water Manage.*, vol. 222, pp. 92–104, Aug. 2019, doi: 10.1016/j.agwat.2019.05.041.

[16] C. Cammalleri and J. Vogt, "On the role of Land Surface Temperature as proxy of soil moisture status for drought monitoring in Europe," *Remote Sens.*, vol. 7, no. 12, pp. 16849–16864, Dec. 2015, doi: 10.3390/rs71215857.

[17] P. Wang, J. Gong, and X. Li, "Vegetation temperature condition index and its application for drought monitoring," *Geomatics Inf. Sci. Wuhan Univ.*, vol. 26, no. 5, pp. 412–418, May 2001, doi: 10.13203/j.whugis2001.05.007.

[18] N. R. Patel, B. R. Parida, V. Venus, S. K. Saha, and V. K. Dadhwal, "Analysis of agricultural drought using vegetation temperature condition index (VTCI) from Terra/MODIS satellite data," *Environ. Monitor. Assessment*, vol. 184, no. 12, pp. 7153–7163, Dec. 2012, doi: 10.1007/s10661-011-2487-7.

[19] W. Sun et al., "Using the vegetation temperature condition index for time series drought occurrence monitoring in the Guanzhong Plain, PR China," *Int. J. Remote Sens.*, vol. 29, nos. 17–18, pp. 5133–5144, Sep. 2008, doi: 10.1080/01431160802036557.

[20] X. Zhou, P. Wang, K. Tansey, S. Zhang, H. Li, and L. Wang, "Developing a fused vegetation temperature condition index for drought monitoring at field scales using Sentinel-2 and MODIS imagery," *Comput. Electron. Agricult.*, vol. 168, Jan. 2020, Art. no. 105144, doi: 10.1016/j.compag.2019.105144.

[21] Z. Wan, P. Wang, and X. Li, "Using MODIS Land Surface Temperature and normalized difference vegetation index products for monitoring drought in the southern Great Plains, USA," *Int. J. Remote Sens.*, vol. 25, no. 1, pp. 61–72, Jan. 2004, doi: 10.1080/0143116031000115328.

[22] J. Khan, P. Wang, Y. Xie, L. Wang, and L. Li, "Mapping MODIS LST NDVI imagery for drought monitoring in Punjab Pakistan," *IEEE Access*, vol. 6, pp. 19898–19911, 2018, doi: 10.1109/ACCESS.2018.2821717.

[23] R. Ullah, J. Khan, I. Ullah, F. Khan, and Y. Lee, "Assessing impacts of flood and drought over the Punjab region of Pakistan using multi-satellite data products," *Remote Sens.*, vol. 15, no. 6, p. 1484, Mar. 2023, doi: 10.3390/rs15061484.

[24] X. Zhu et al., "A novel framework to assess all-round performances of spatiotemporal fusion models," *Remote Sens. Environ.*, vol. 274, Jun. 2022, Art. no. 113002, doi: 10.1016/j.rse.2022.113002.

[25] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006, doi: 10.1109/TGRS.2006.872081.

[26] X. Zhu, J. Chen, F. Gao, X. Chen, and J. G. Masek, "An enhanced spatial and temporal adaptive reflectance fusion model for complex heterogeneous regions," *Remote Sens. Environ.*, vol. 114, no. 11, pp. 2610–2623, Nov. 2010, doi: 10.1016/j.rse.2010.05.032.

[27] X. Zhu, E. H. Helmer, F. Gao, D. Liu, J. Chen, and M. A. Lefsky, "A flexible spatiotemporal method for fusing satellite images with different resolutions," *Remote Sens. Environ.*, vol. 172, pp. 165–177, Jan. 2016, doi: 10.1016/j.rse.2015.11.016.

[28] M. Schmidt, R. Lucas, P. Bunting, J. Verbesselt, and J. Armston, "Multi-resolution time series imagery for forest disturbance and regrowth monitoring in Queensland, Australia," *Remote Sens. Environ.*, vol. 158, pp. 156–168, Mar. 2015, doi: 10.1016/j.rse.2014.11.015.

[29] L. V. Oldoni et al., "Extraction of crop information through the spatiotemporal fusion of OLI and MODIS images," *Geocarto Int.*, vol. 37, no. 25, pp. 8336–8360, Dec. 2022, doi: 10.1080/10106049.2021.2000648.

[30] A. Jarihani, T. McVicar, T. Van Niel, I. Emelyanova, J. Callow, and K. Johansen, "Blending Landsat and MODIS data to generate multispectral indices: A comparison of 'index-then-blend' and 'blend-then-index' approaches," *Remote Sens.*, vol. 6, no. 10, pp. 9213–9238, Sep. 2014, doi: 10.3390/rs6109213.

[31] F. Tian, Y. Wang, R. Fensholt, K. Wang, L. Zhang, and Y. Huang, "Mapping and evaluation of NDVI trends from synthetic time series obtained by blending Landsat and MODIS data around a coalfield on the loess Plateau," *Remote Sens.*, vol. 5, no. 9, pp. 4255–4279, Sep. 2013, doi: 10.3390/rs5094255.

[32] X. Chen, M. Liu, X. Zhu, J. Chen, Y. Zhong, and X. Cao, "'Blend-then-index' or 'index-then-blend': A theoretical analysis for generating high-resolution NDVI time series by STARFM," *Photogramm. Eng. Remote Sens.*, vol. 84, no. 2, pp. 65–73, Feb. 2018, doi: 10.14358/pers.84.2.65.

[33] H. Tian et al., "A deep learning framework under attention mechanism for wheat yield estimation using remotely sensed indices in the Guanzhong Plain, PR China," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 102, Oct. 2021, Art. no. 102375, doi: 10.1016/j.jag.2021.102375.

[34] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Jan. 2001, doi: 10.1023/a:1010933404324.

[35] M. Belgiu and L. Drăguţ, "Random forest in remote sensing: A review of applications and future directions," *ISPRS J. Photogramm. Remote Sens.*, vol. 114, pp. 24–31, Apr. 2016, doi: 10.1016/j.isprsjprs.2016.01.011.

[36] C. Hutengs and M. Vohland, "Downscaling Land Surface Temperatures at regional scales with random forest regression," *Remote Sens. Environ.*, vol. 178, pp. 127–141, Jun. 2016, doi: 10.1016/j.rse.2016.03.006.

[37] H. Ebrahimy and M. Azadbakht, "Downscaling MODIS Land Surface Temperature over a heterogeneous area: An investigation of machine learning techniques, feature selection, and impacts of mixed pixels," *Comput. Geosci.*, vol. 124, pp. 93–102, Mar. 2019, doi: 10.1016/j.cageo.2019.01.004.

[38] W. P. Kustas, J. M. Norman, M. C. Anderson, and A. N. French, "Estimating subpixel surface temperatures and energy fluxes from the vegetation index-radiometric temperature relationship," *Remote Sens. Environ.*, vol. 85, no. 4, pp. 429–440, Jun. 2003, doi: 10.1016/S0034-4257(03)00036-1.

[39] Y. Ma et al., "Estimation of daily evapotranspiration and irrigation water efficiency at a Landsat-like scale for an arid irrigation area using multi-source remote sensing data," *Remote Sens. Environ.*, vol. 216, pp. 715–734, Oct. 2018, doi: 10.1016/j.rse.2018.07.019.

[40] X. Bei et al., "Estimation of daily terrestrial latent heat flux with high spatial resolution from MODIS and Chinese GF-1 data," *Sensors*, vol. 20, no. 10, p. 2811, May 2020, doi: 10.3390/s20102811.



Yue Zhang received the B.S. degree in geographic information science from Shanxi Agricultural University, Jinzhong, China, in 2018. She is currently pursuing the Ph.D. degree in computer science and technology with China Agricultural University, Beijing, China.

Her research interests focus on quantitative remote sensing and its application in agriculture.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.



Pengxin Wang received the B.Sc. and M.Sc. degrees in agronomy from Northwestern Agricultural University (NWAU), Xianyang, China, in 1988 and 1991, respectively, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2001.

From 1991 to 1998, he was a Faculty Member with NWAU. He has been a Professor with China Agricultural University, Beijing, China, since 2003. He teaches remote sensing at the undergraduate and graduate levels. His main research interests include the application of quantitative remote sensing in agriculture.



Kevin Tansey received the B.S. degree in physical geography from the University of Sheffield, Sheffield, U.K., in 1995, and the Ph.D. degree in remote sensing and physical geography from the University of Leicester, Leicester, U.K., in 1999.

He has held research positions at Swansea University, Swansea, U.K.; Friedrich-Schiller University, Jena, Germany; and the Joint Research Centre, European Commission, Ispra, Italy. He is currently a Professor of remote sensing with the School of Geography, Geology and the Environment, University of Leicester. His research interests include remote sensing of land surface dynamics in relation to vegetation growth and disturbance.



Mingqi Li received the B.S. degree in computer science and technology from Hebei Agricultural University, Cangzhou, China, in 2018. He is currently pursuing the Ph.D. degree in computer science and technology with China Agricultural University, Beijing, China.

His research interests include quantitative remote sensing and its application in agriculture.



Fengwei Guo received the B.S. degree in computer science and technology from Jiangnan University, Wuxi, China, in 2022. He is currently pursuing the Ph.D. degree in computer science and technology with China Agricultural University, Beijing, China.

His research interests include the application of quantitative remote sensing in agriculture.



Junming Liu received the B.Sc. degree in agricultural meteorology, the M.Sc. degree in information management, and the Ph.D. degree in soil science from China Agricultural University, Beijing, China, in 1993, 1996, and 2003, respectively.

Since 2007, he has been an Associate Professor with China Agricultural University. His main research interests include the geographic information system and the application of remote sensing in agriculture.



Shuyu Zhang received the B.Sc. degree in electronic engineering and computer science from Northwestern Polytechnical University, Xi'an, China, in 1990.

He is currently a Researcher at the Shaanxi Provincial Meteorological Bureau, Xi'an. His main research interests include the applications of remote sensing in agriculture and meteorology.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:34:44 UTC from IEEE Xplore. Restrictions apply.